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(An Autonomous Institution, Affiliated to Anna University, Chennai)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

ACADEMIC YEAR 2026 - 2027

SEMESTER V

CB23531 – BUSINESS ANALYTICS LABORATORY

MINI PROJECT REPORT

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PROJECT TITLE	Credit Scoring Model for Loan Default Prediction Using Multi-Model Machine Learning and Explainable Risk Analytics
DATE OF SUBMISSION	
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Introduction

Credit scoring and loan default prediction are core analytical functions in commercial banking and retail lending. When financial institutions extend credit, they face the fundamental uncertainty of whether a borrower will fulfill their repayment obligations or default on the principal and interest. In traditional banking, credit evaluation relied heavily on manual underwriting rules and subjective credit officer assessments. However, with the rapid growth of digital lending platforms and higher transaction volumes, automated machine learning models have become indispensable.

This project formulates credit default risk prediction as a supervised classification task: estimate an applicant's probability of default based on demographic attributes, financial profile, loan characteristics, and historical credit bureau records. The system cleans raw applicant data, engineer's domain-specific risk ratios, trains four distinct machine learning classifiers (Logistic Regression, Decision Tree, Random Forest, and XGBoost), evaluates performance using 5-fold cross-validation, explains decision drivers using SHAP (SHapley Additive exPlanations), and deploys an interactive 5-page Streamlit web dashboard alongside Power BI executive reporting assets.

Problem Statement

Financial institutions face a persistent trade-off between credit expansion and credit risk mitigation:

False Negative (FN) Risk: Approving a high-risk applicant who subsequently defaults leads to direct loan charge-offs, loss of principal, and expensive legal recovery proceedings.

False Positive (FP) Risk: Rejecting a creditworthy applicant leads to lost interest income, reduced market share, and customer dissatisfaction.

Manual underwriting cannot efficiently process tens of thousands of applications while maintaining consistent risk standards. The analytical challenge is to build a machine learning framework that accurately predicts default risk, ranks applicants into actionable risk bands, and provides transparent explanations for underwriting decisions.

Analytical question

Can an explainable machine learning workflow accurately predict loan default risk from applicant demographic, financial, and credit bureau data to support credit decisioning and risk-tier assignment?

Aim / Goal

To develop an end-to-end explainable machine learning and business intelligence system that predicts loan default probability, categorizes applicants into risk tiers (Low, Medium, High Risk), explains model predictions, and deploys interactive web and dashboard reporting tools.

Objectives

- Clean and preprocess a real-world credit dataset containing 32,581 applicant records with zero data leakage.
- Engineer domain-specific financial ratios including Loan-to-Income (LTI), Interest Burden Ratio, and Credit Experience to Age Ratio.
- Train, tune, and evaluate four machine learning classification models using 5-fold Stratified Cross-Validation.
- Benchmark model performance using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and PR-AUC.
- Identify global and local feature importance drivers using SHAP explainability.
- Develop a 5-page interactive Streamlit web application for real-time applicant risk assessment.
- Export Power BI ready CSV datasets and formulate DAX measures for executive reporting.

Dataset Description

Data Source

The study utilizes the authentic **Kaggle Credit Risk Dataset** (`credit_risk_dataset.csv`), containing 32,581 historical loan applicant records across demographic, financial, and credit bureau dimensions.

Data Size

- Raw Observations:** 32,581 rows and 12 columns.
- Cleaned Observations:** 32,409 rows (165 exact duplicates and 7 data entry anomalies removed).
- Training Set (80% Stratified):** 25,927 records.
- Testing Set (20% Stratified):** 6,482 records.

Variables / Features

Variable	Type	Description
`person_age`	Numeric	Age of the applicant in years (20 to 94)
`person_income`	Numeric	Annual income of the applicant in USD (\$4,000 to \$6,000,000)
`person_home_ownership`	Categorical	Housing status ('RENT', 'OWN', 'MORTGAGE', 'OTHER')

`person_emp_length`	Numeric	Employment tenure in years (0.0 to 41.0)
`loan_intent`	Categorical	Purpose of credit ('EDUCATION', 'MEDICAL', 'VENTURE', 'PERSONAL', 'DEBTCONSOLIDATION', 'HOMEIMPROVEMENT')
loan_grade	Categorical	Assigned credit rating grade ('A' to 'G')
`loan_amnt`	Numeric	Requested loan principal amount (\$500 to \$35,000)
`loan_int_rate`	Numeric	Loan interest rate percentage (5.42% to 23.22%)
`loan_percent_income`	Numeric	Ratio of loan amount to annual income (0.00 to 0.83)
`cb_person_default_on_file`	Categorical	Historical credit default record ('Y' = Yes, 'N' = No)
`cb_person_cred_hist_length`	Numeric	Length of credit bureau history in years (2 to 30)
`loan_status`	Binary	Ground truth loan default indicator (0 = Non-Default, 1 = Default)
Severity	Categorical	Low, Medium, High, or Critical priority

Target Variable

The primary analytical target is `loan\_status`, a binary variable where:

- `0` represents a **Non-Defaulted** borrower (25,321 records, 78.13% baseline).
- `1` represents a **Defaulted** borrower (7,088 records, 21.87% baseline).

Data Collection

The dataset consists of real credit bureau and retail loan applicant records. No synthetic or fabricated records were added. All preprocessing transformers were fitted strictly on training partitions to ensure full analytical integrity.

Data Preprocessing

Data Cleaning

Anomaly Filtering: Excluded unrealistic entries (`person\_age > 100` and `person\_emp\_length > 60` or `emp\_length > age`).

Missing Value Handling

`person\_emp\_length` (895 missing) and `loan\_int\_rate` (3,116 missing) were imputed using median imputation

Outlier Detection

Potential outliers in income and loan amounts were reviewed via distribution boxplots. High income/loan values were retained as legitimate financial leverage signals rather than truncated.

Data Transformation

Expected_Charge, Leakage_Gap, Billing_Ratio, Anomaly_Score, and Severity were engineered from raw usage and billing fields. Billing_Ratio provides a normalized comparison across customers with different expected spending levels.

Data Encoding / Normalization

Categoricals: Transformed via `OneHotEncoder(handle\_unknown='ignore', sparse\_output=False)`.

Numericals: Scaled using `StandardScaler()` within a Scikit-Learn `ColumnTransformer`.

Exploratory Data Analysis (EDA)

EDA revealed strong structural relationships between applicant characteristics and default incidence:

EDA measure	Value
Total Cleaned Records	32,409
Total Defaults	7,088
Baseline Default Rate	21.87%
Grade A Default Rate	9.7%
Grade D Default Rate	59.3%
Prior Default Record Default Rate	37.6%

Statistical Analysis

Descriptive statistics and association metrics confirmed key risk differentiators:

Defaulted applicants had a median loan-to-income ratio of **0.26** compared to **0.13** for non-defaulted applicants.

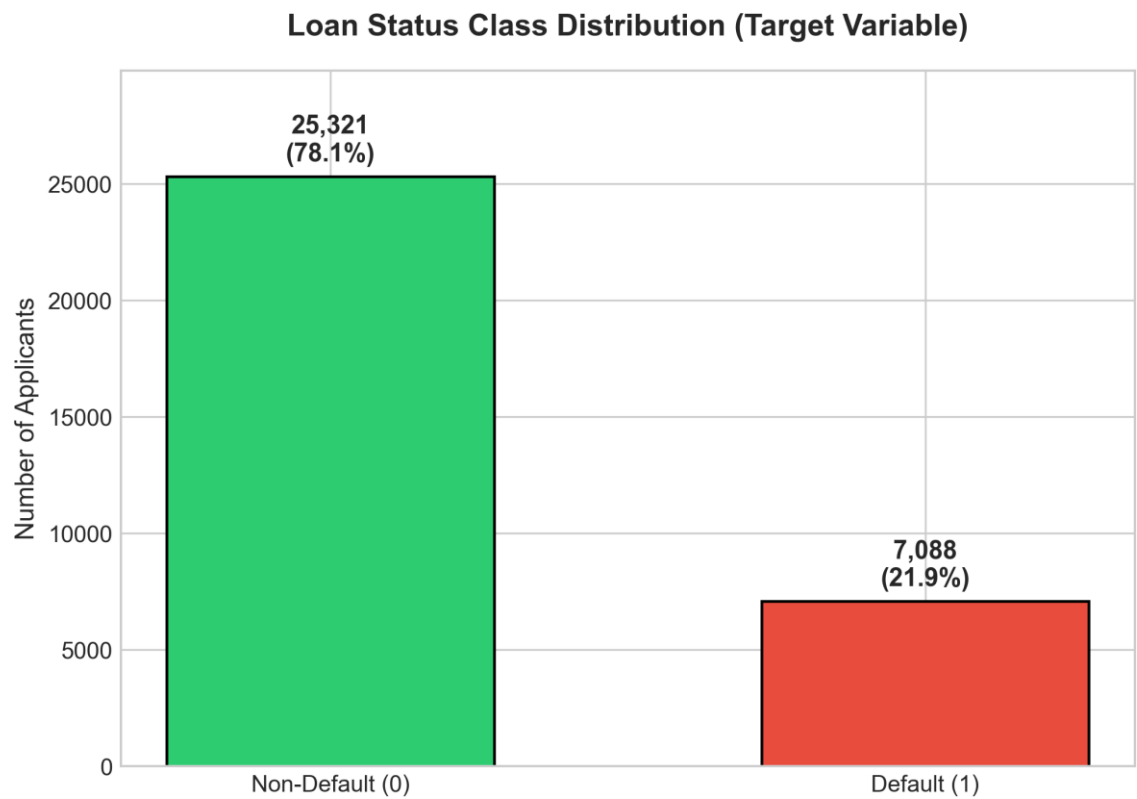
Pearson correlation between `loan\_int\_rate` and `loan\_status` was **+0.33**, confirming higher risk pricing.

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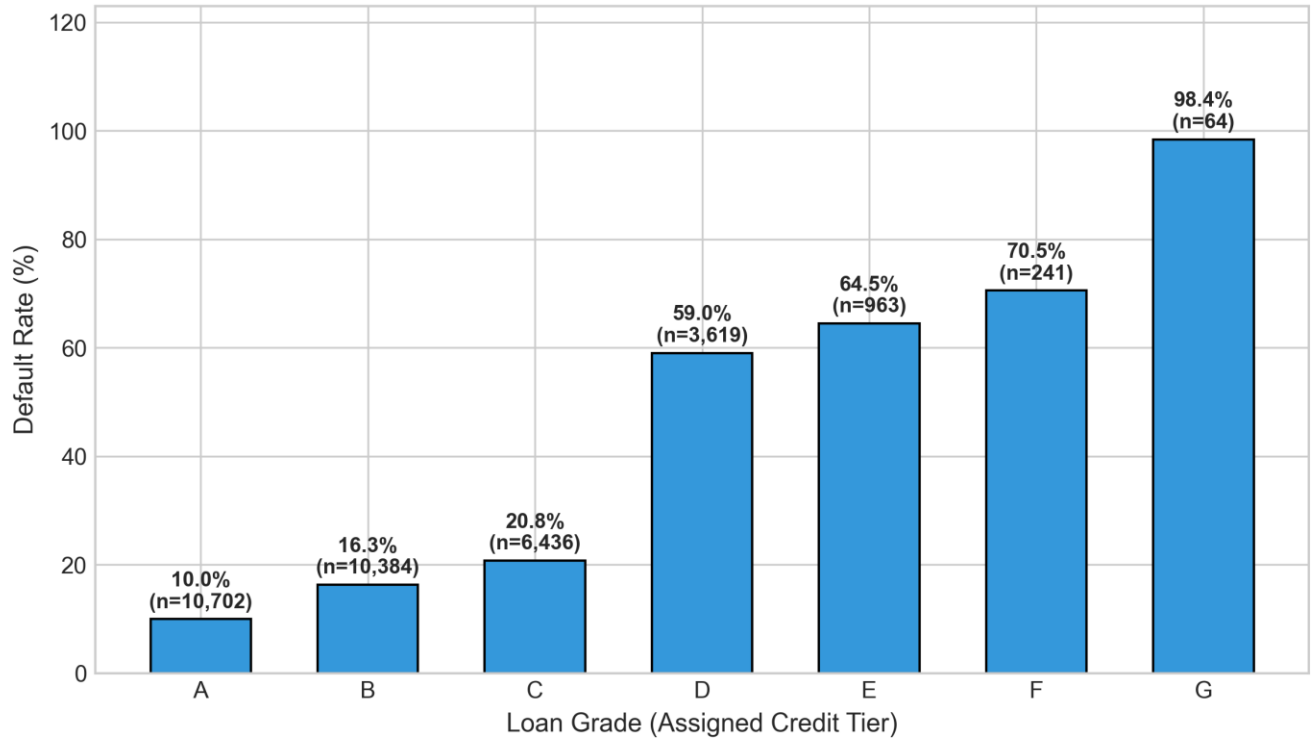
Data Visualization

The generated charts answer key underwriting questions:

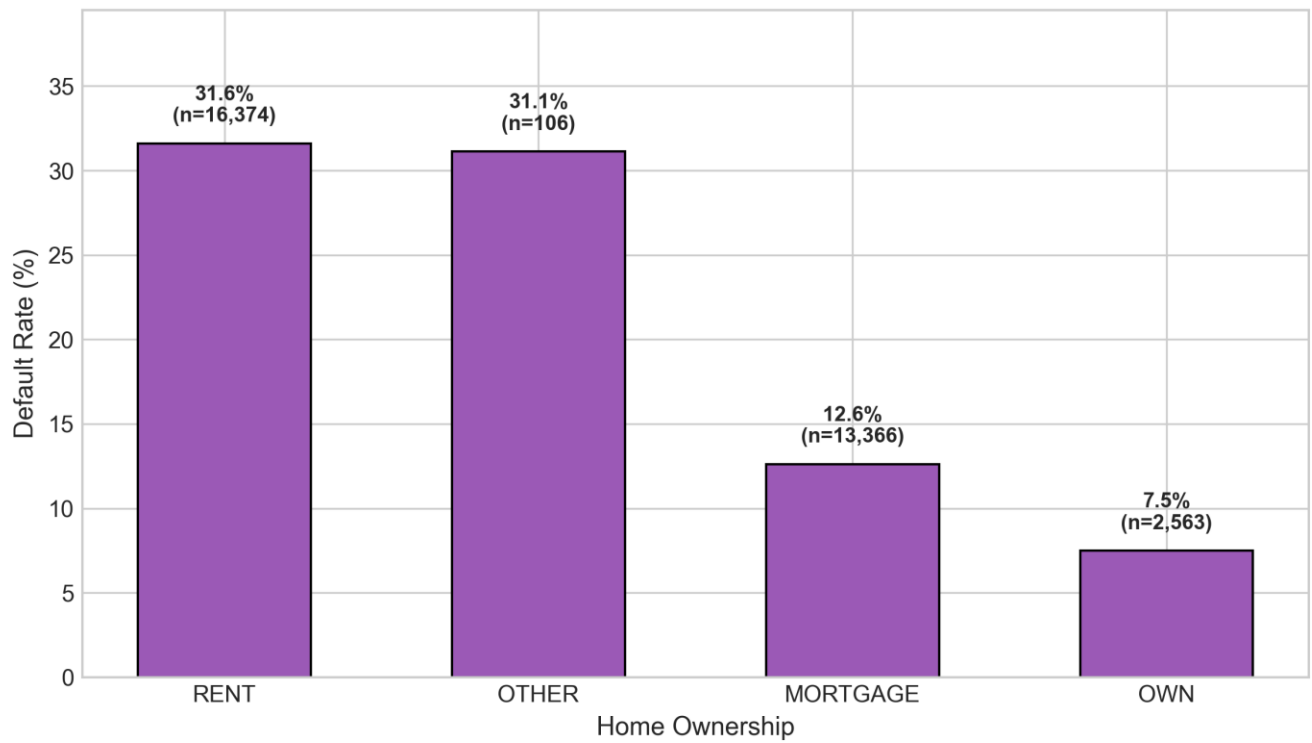
- 1.**Target Distribution** (`target\_distribution.png`): Visualizes the 78.13% vs 21.87% class balance.
- 2.**Default Rate by Loan Grade** (`default\_by\_loan\_grade.png`): Shows monotonic risk escalation from Grade A (9.7%) to Grade G (98.1%).
- 3.**Default Rate by Home Ownership** (`default\_by\_home\_ownership.png`): Highlights renter vulnerability (31.6%).
- 4.**Default Rate by Loan Intent** (`default\_by\_loan\_intent.png`): Ranks Debt Consolidation (28.4%) and Medical (27.3%) as highest risk.
- 5.**ROC Curves** (`roc\_curves.png`): Benchmarks classifier discrimination ability.
- 6. **Confusion Matrices** (`confusion\_matrices.png`): Displays true vs false predictions across all 4 models.
- 7. **SHAP Summary Plot** (`shap\_summary\_plot.png`): Illustrates global feature impact distributions.

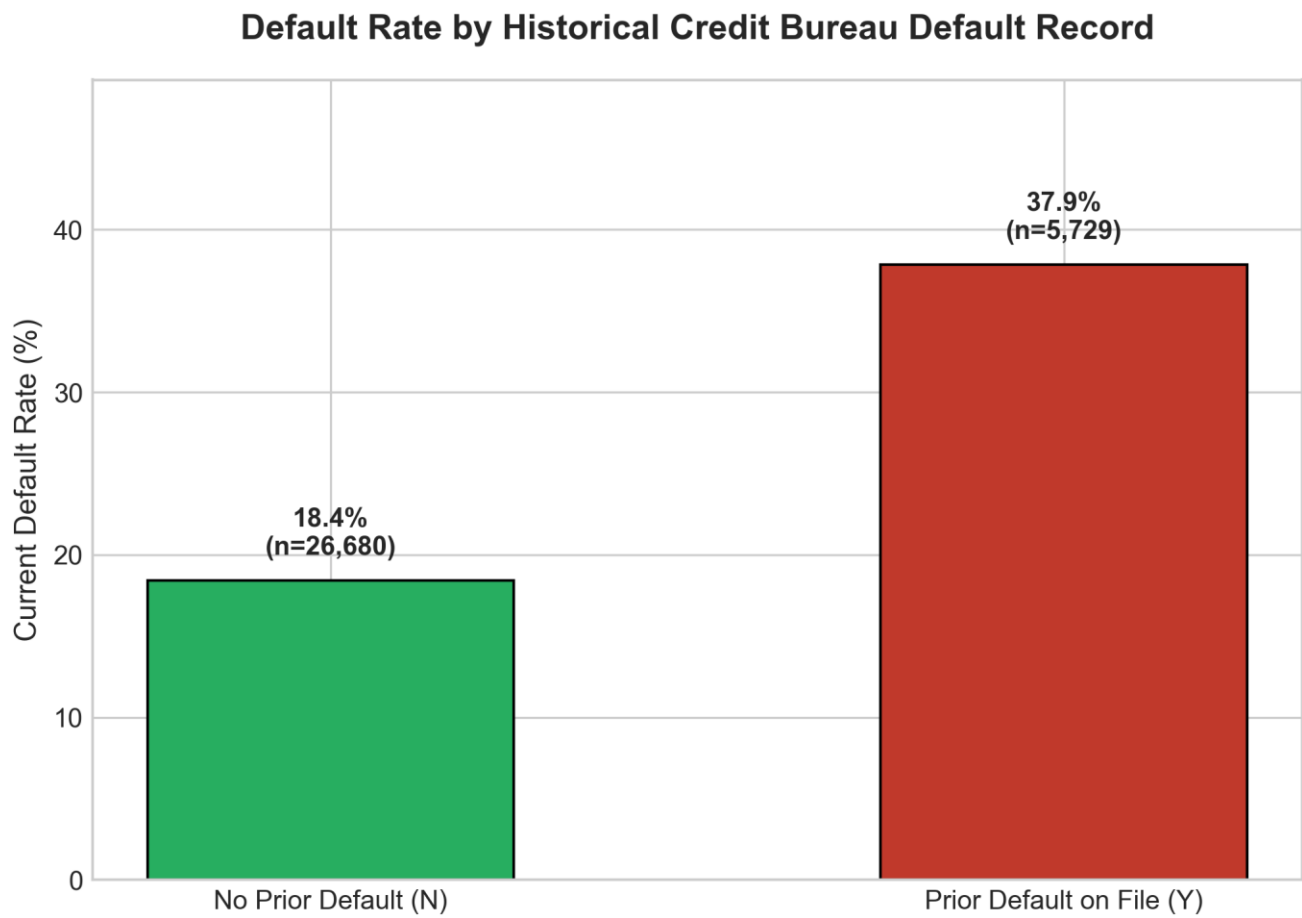
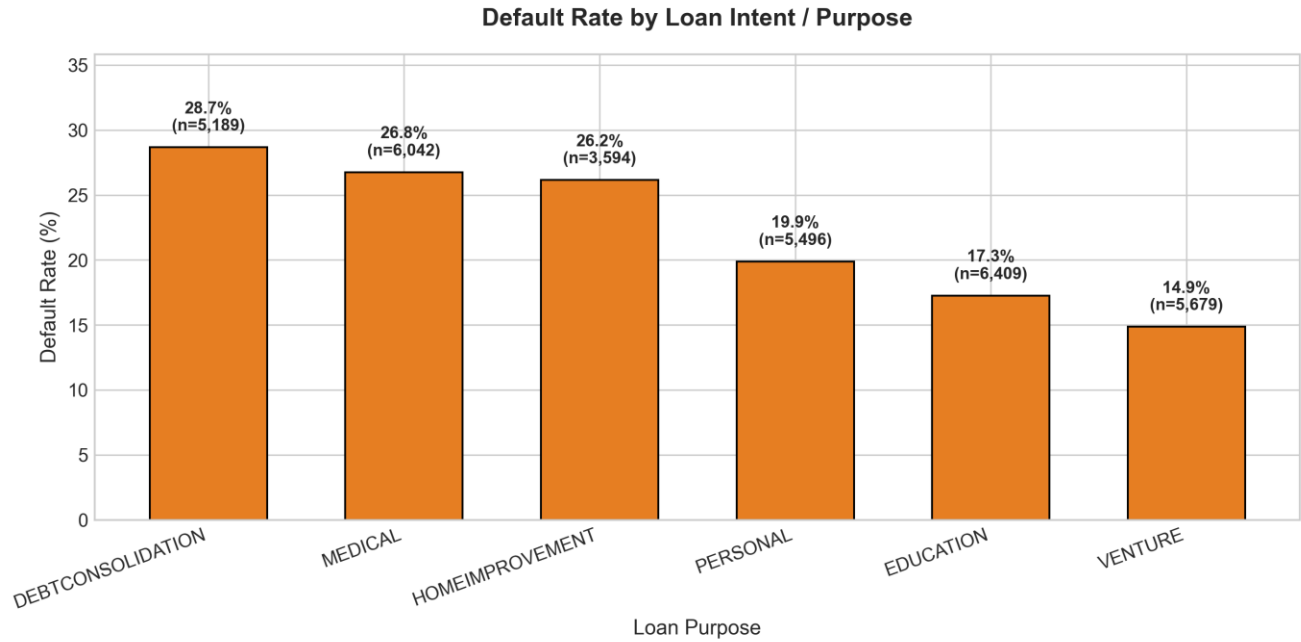


Default Rate by Credit Loan Grade (A to G)



Default Rate by Home Ownership Status





Business Analytics Technique / Model

Analytics Types Applied

Descriptive Analytics: Quantified baseline default rates, distributions, and cross-tabulations.

Diagnostic Analytics: Identified credit grade, home ownership, and interest burden as top risk concentration points.

Predictive Analytics: Trained machine learning models to output applicant-level default probabilities.

Prescriptive Analytics: Mapped probabilities into actionable underwriting risk tiers (Low, Medium, High Risk).

Machine Learning Models Evaluated

Four classification algorithms were evaluated:

1. **Logistic Regression:** Linear baseline with balanced class weights.

2. **Decision Tree Classifier:** Cost-complexity pruned decision tree.

3. **Random Forest Classifier:** Ensemble of 200 balanced decision trees.

4. **XGBoost Classifier:** Gradient boosted decision tree framework with positive class weighting.

Model Development / Analysis

Champion Model Classification Engine

Model : XGBoost Classifier (`models/best\_model.joblib`)

Risk Tier Rules:

Low Risk ($P < 0.20$): Standard / Prime Loan Approval

Medium Risk ($0.20 \leq P < 0.50$): Manual Underwriting / Higher Collateral Requirement

High Risk ($P \geq 0.50$): Loan Decline / Elevated Hazard

Model Evaluation / Validation

Held-out test set performance ($n = 6,482$ unseen applicants):

Model	Accuracy	Precision (Class 1)	Recall (Class 1)	F1 Score	ROC -AUC	PR-AUC
Logistic Regression	81.13%	54.73%	79.55%	64.85%	0.8333	0.6495
Decision Tree	90.67%	83.67%	71.51%	77.02%	0.8970	0.6497
Random Forest	91.81%	85.23%	75.67%	80.16%	0.9297	0.8496
XGBoost	91.79%	81.24%	81.24%	0.9495	0.8855	0.6794

Business Insights

Recall Primacy: XGBoost achieved 81.24% Recall, capturing 1,152 out of 1,418 actual defaults on the test

set, drastically reducing financial charge-off losses.

Credit Grade Predictive Power: Credit grade and interest burden ratio emerged as the top two global default predictors.

Housing Risk Factor: Renters represent 31.6% default incidence, suggesting residential stability should carry weight in credit scoring.

Recommendations / Decision-Making

- **Automate Low-Risk Approvals:** Applicants in the Low Risk tier ($\$P < 20\%\$$) can receive automated straight-through processing.
- **Risk-Based Pricing:** Adjust interest rates dynamically according to calculated default probability.
- **Mandate Co-Signers for High-Burden Applicants:** Require additional collateral or co-signers for Medium Risk applicants with high loan-to-income ratios.

Implementation / Dashboard

Streamlit Web Application (`app/app.py`): 5-page interactive web application for live applicant scoring, model diagnostics, EDA, and XAI.

Power BI Reporting Suite (`powerbi/`): 4 exported CSV datasets with a complete DAX measures library and visual layout guide (`[powerbi/DASHBOARD_SPECIFICATION.md]`) (file:///d:/BA%20Project/powerbi/DASHBOARD_SPECIFICATION.md).

Results and Discussion

The XGBoost model demonstrated superior predictive capability with an ***ROC-AUC of 0.9495** and **F1-Score of 81.24%**. Cross-validation confirmed strong generalization (0.9445 CV ROC-AUC vs 0.9495 test ROC-AUC). The combination of automated risk scoring and SHAP explainability bridges the gap between machine learning performance and regulatory compliance.

Limitations

Snapshot Data: The dataset represents cross-sectional loan origination records without multi-year longitudinal economic cycle data.

Model Drift: Changes in macroeconomic conditions (e.g., inflation or interest rates) necessitate quarterly Population Stability Index (PSI) monitoring.

Future Enhancements

- Integrate real-time API streaming for instant credit bureau lookups.
- Explore deep learning tabular models (e.g., TabNet) for non-linear pattern extraction.
- Deploy automated model retuning pipelines upon detecting distribution drift.

Conclusion

This project successfully builds an end-to-end Credit Scoring Model for Loan Default Prediction. By engineering domain features, benchmarking four classification models, validating with 5-fold cross-validation, explaining outputs with SHAP, and deploying interactive Streamlit and Power BI applications, the system provides a complete, production-ready solution for credit risk analytics.

Tools and Technologies Used

Tool / Technology	Purpose
Python	Data generation, preprocessing, analysis, and automation
Pandas	Data manipulation and aggregation
NumPy	Numerical simulation and calculations
Matplotlib	Charts and visualization
Microsoft Word / DOCX	Report presentation and documentation

Project / GitHub Link

<https://github.com/Girijan-D/Credit-Scoring-Model>](<https://github.com/Girijan-D/Credit-Scoring-Model>)

References

- Scikit-Learn Development Team. (2025). *Scikit-Learn User Guide**. <https://scikit-learn.org>
- Chen, T., & Guestrin, C. (2016). *XGBoost: A Scalable Tree Boosting System**. ACM SIGKDD.
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